

Level 2 - Searching Core Problems

1. First Occurance

```
nums = [1, 3, 3, 3, 3, 8]
target = 3

def first_occurance(nums):
    left = 0
    right = len(nums) - 1
    ans = -1
    while left <= right:
        mid = (left + right) // 2

        if nums[mid] == target:
            ans = mid
            right = mid - 1

        elif nums[mid] < target:
            left = mid + 1

        else:
            right = mid - 1

    return ans

print(first_occurance(nums))
```

2. Last Occurance

```
nums = [1, 3, 3, 3, 3, 8]
target = 3

def last_occurance(nums):
    left = 0
    right = len(nums) - 1
    ans = -1
    while left <= right:
        mid = (left + right) // 2

        if nums[mid] == target:
            ans = mid
            left = mid + 1

        elif nums[mid] < target:
            left = mid + 1

        else:
            right = mid - 1

    return ans

print(last_occurance(nums))
```

3. Lower bound

```
nums = [1, 3, 3, 5, 7]
target = 4

def lower_bound(nums):
    left = 0
    right = len(nums) - 1
    ans = len(nums)

    while left <= right:
        mid = (left + right) // 2

        if nums[mid] >= target:
            ans = mid
            right = mid - 1

        else:
            left = mid + 1

    return ans

print(lower_bound(nums))
```

4. Upper Bound

```
nums = [1, 2, 2, 2, 4, 5]
target = 2

def upper_bound(nums):
    left = 0
    right = len(nums) - 1
    ans = len(nums)

    while left <= right:
        mid = (left + right) // 2
        if nums[mid] > target:
            ans = mid
            right = mid - 1

        else:
            left = mid + 1

    return ans

print(upper_bound(nums))
```